

Considering transient state in interconnected networks during fault for coordination of directional overcurrent relays

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ABSTRACT

The object of optimal coordination in networks is to obtain the minimum operating time (OT) of directional overcurrent relays and to maintain the coordination time interval. Coordination of relays in the steady state causes miscoordination in practice. Conventional methods have some drawbacks such as ignoring the exact formula of the relay, using the simplified model of the network elements (ignoring complete models and non-linear elements), and disregarding backup relays chain and fault current redistribution during topology transient change. In this study, a coordination method is proposed that considers the transient state by simulating the network using a complete model of the elements and precise formulas. Network simulation was performed in three stages in Simulink during the coordination. A new method is also proposed to obtain and correctly indicate the backup relays OTs incorporating the topology transient change. Initially, using one conventional method, the IEEE 8-bus and IEEE 30-bus networks were coordinated. The results show 23 and 94 mismatches, respectively. Moreover, since relays' OTs show some errors, they were then coordinated using the proposed method. Consequently, all mismatches were eliminated and error-free OTs of relays were obtained. This study confirms that the proposed method has more practical results compared to conventional methods.

1. Introduction

Directional Overcurrent relays (DOCRs) are used in all interconnected power networks. Design criteria for system protection include selectivity, reliability, speed of operation, and sensitivity. The selectivity goal in protection systems is achieved through the coordination of relays. The goal behind the relay coordination is to determine Time Multiplier Setting (TMS) and Plug Multiplier Setting (PMS) of all relays. Recently, it is feasible to select the relay characteristic curve that is determined for every relay in coordination [1].

Some miscoordination may occur due to ignoring factors such as DOCR exact formula, instantaneous fault currents, network available complete model, nonlinear elements of the network, and **topology transient change**. Ignoring the transient state of the network is the main factor in this regard. In the following, some studies on the prevention of miscoordination in the literature are discussed. Incorporating network different topologies is the main issue of some papers. There are two main approaches for coordination of relays in static topologies: 1) coordination of relays for all possible static topologies during optimization [2,3] and 2) adaptive coordination [4]. When changing the topology of the network, neglecting the Fault Current Redistribution

(FCR) leads to miscoordination. A new constraint for minimization of the operating time gap between primary relays at near and far ends of the line to prevent FCR is proposed in Singh and Panigrahi [5], Singh et al. [6]. A phase fault protection scheme is introduced in Sharma and Panigrahi [7] by imposing some additional constraints on the conventional relay coordination problem to prevent undesired relay tripping. Changing the shape of the relay characteristic curve to consider FCR is studied in Enríquez and Martínez [8]. In these papers, permanent fault currents are used while ignoring the instantaneous fault currents.

Inappropriate use of the relay Operating Time (OT) calculation formula, as stated in the IEC 60255-151:2009 standard [9], is another miscoordination factor. A comparison of the fault clearing actual time with the calculated OT in the general formula is performed in Efthymiadis et al. [10]. The way to implement the IEC standard formula for calculating the relays OT is shown in Wang and Bollen [11]. However, these calculations are based on steady-state fault currents, before and after tripping the first-main relay to obtain accurate OT of a relay. So, it is required to use the instantaneous fault currents.

Other important factors of miscoordination are disregarding the Current Transformers (CTs) saturation, circuit breakers and power transformers effects, and the complete model of the network elements,

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which have a direct impact on instantaneous fault currents. It has been reported that there would be a miscoordination when using an ideal model of power source instead of the actual model [12]. By evaluating one formerly coordinated network in ETAP, it is proved that analyzing networks in the static condition is inappropriate [13]. The effect of current saturation on digital overcurrent relays operation is investigated in Tziouvaras et al. [14]. In the literature, CT saturation has not been much considered during relays optimal coordination. The circuit breaker interrupts the current of each phase at different times. Power transformers can also cause fault current distortion. To the best of our knowledge, these factors have not yet been considered in relays coordination.

In all the papers cited, only one factor related to miscoordination is considered. Moreover, the related works do not apply the complete model of network elements during coordination and ignore the transient state of the network [5–7]. The complete models of network elements are widely available in the time-domain simulators like Simulink. Therefore, the network simulation in the time domain should be regarded during coordination, which is the major shortcoming of conventional coordination methods. In this paper, the instantaneous fault currents and the DOCRs exact formula are used to obtain the OT of DOCRs. To consider FCR, the simulation is performed in three stages. Considering the problems involved in the common method of expressing the backup relays OT, a new method is needed to deal with this issue. Here, the Particle Swarm Optimization (PSO) is applied as the optimization algorithm. The main contributions of this paper are as follows:

1. Obtaining errorless operating time of relays by considering the transient states
2. Elimination of miscoordination actual factors by network complete simulation during the coordination
3. The correct expression of the backup relays OT while the fault current redistribution is considered

2. Problems leading to miscoordination

The relays coordination is majorly based on the fault current calculation. Conventionally, two methods exist for the calculation of the fault currents:

1. Analytical method: In this method, to obtain the buses voltage and the lines current in steady state, first, the load flow calculations are performed. Then, the Thevenin equivalent circuit is computed in the fault location by calculating the network impedance matrix. The fault is then applied to the network to calculate its effect on currents and voltages. Finally, fault currents are calculated by applying changes from the faults to the load flow results [15].
2. Modeling method: Steady-state fault currents can be determined by modeling the network in suitable software once in the first step of coordination.

The shortcomings of these methods, along with other problems, lead to miscoordination and are listed as follows:

1. Ignoring the relay exact formula
2. Using the simplified model of the network elements
 - (a) Ignoring the complete model of the network elements
 - (b) Ignoring the nonlinear elements of the network
3. Fault current redistribution during topology transient change
4. Disregarding backup relays chain

Ignoring the instantaneous fault current can lead to errors in OT of relays and consequently on coordination.

2.1. Ignoring the relay exact formula

Transient states affect the operation of the relays. Disregarding the transient currents is one of the shortcomings of relays conventional coordination [13].

Conventional methods use the following formula, which is based on the stable fault current:

$$t = TMS \left(\frac{k}{(I_{fault}/I_{pickup})^\alpha - 1} + c \right) \quad (1)$$

where, I_{fault} is the steady-state fault current and I_{pickup} is the current setting of the relay; α , k , and c are the constant coefficients of the relay characteristic curve. The calculated OT using this formula is not accurate because it ignores the instantaneous currents. To consider transient currents, the exact formula of the relay must be used. Based on the IEC standard [9], the OT of a relay is as follows:

$$t(G) = TMS \left(\frac{k}{(G/G_s)^\alpha - 1} + c \right) \quad (2)$$

$$\int_0^{T_0} \frac{1}{t(G)} dt = 1 \quad (3)$$

where, T_0 is the relay OT; G is instantaneous fault current; G_s and TMS are current and time setting of the relay respectively. Therefore, all transient values of the fault currents are important and have significant effect on OT.

Fig. 1 shows the IEEE 8-bus network [16]. Fig. 2 illustrates the effective and transient fault current of R1 for a fault near it in the L1 line. The steady-state fault current in this figure is commonly used for relays coordination. Consequently, using this current leads to the computation error of OTs and results in miscoordination.

2.2. Using the simplified model of the network elements

The network consists of several nonlinear and complex elements. In the conventional calculations of fault currents, the elements of the network are assumed as linear. Besides, the simplified models of the network elements are used for calculations of fault currents. These factors lead to calculation errors.

2.2.1. Ignoring the complete model of the network elements

The elements in the network can be modeled with a simple or complete equivalent circuit. The conventional methods employ the simple model of the element. As a result, fault currents and OTs are not accurately calculated. The power source is one of the important elements of the network. If the power sources were simulated as ideal voltage sources, their dynamics will be ignored [12].

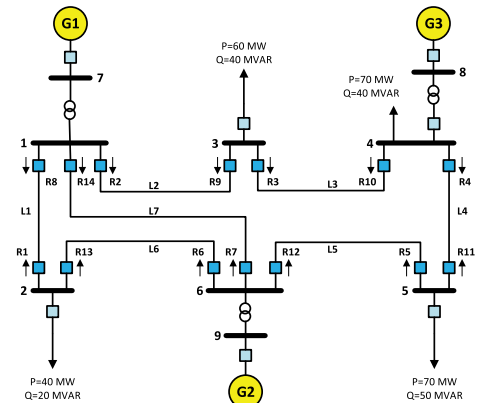


Fig. 1. IEEE 8-bus network.

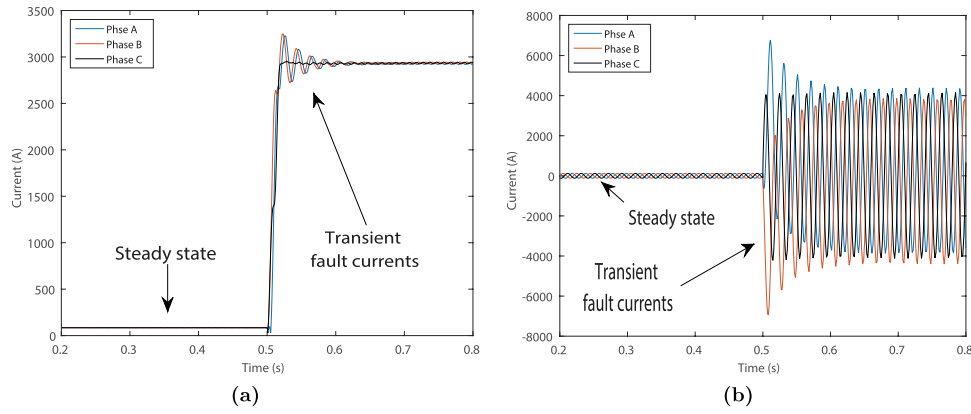


Fig. 2. Transient currents of R1: (a) R1 RMS currents; (b) R1 instantaneous currents.

2.2.2. Ignoring the nonlinear elements of the network

The power transformer is a nonlinear element in the network and maybe saturated during a fault. In the conventional methods, the transformers are modeled with a single reactance, while the saturation of the transformer can be modeled using the complete model.

The CT core may also be saturated when a fault occurs. Saturated CT distorts secondary current. Consequently, the secondary effective current of the CT decreases during the transient period and causes an error in the OTs. The effect of R1 CT core saturation is presented in Fig. 3 for a fault near R1. Fig. 3b shows that the saturated effective current is less than the unsaturated current. As can be seen, the effective current is 3068 A, which decreases to 2680 A when the core is saturated. The CT modeling method is presented in ROMANYUK [17]. The IEEE Std C37.110 standard has introduced some requirements to prevent CT saturation [18]. These requirements generally result in impractically large CTs. Choosing a smaller CT means accepting the saturation. Therefore, CT saturation should be considered in relay coordination. Considering the CT saturation in the procedure of coordination is very complex and is ignored in common methods. However, the transient values of fault currents cannot be ignored.

2.3. Fault current redistribution during topology transient change

The lines in the transmission network are protected by two DOCRs at two ends of the line. When a fault occurs in a line, fault currents flow to the fault location from different paths. When one main relay operates, the network topology changes before the second main relay trips [5,6]. As a result, the fault current is redistributed in the network's new topology. Therefore, two different fault currents occur in the network. The first fault current is the current before a relay trips. The second fault current is the current that flows after a relay trips. In Fig. 1, R7 and R14 are the main DOCRs of L7. R1 and R9 are the backups of

R14. Fig. 4 illustrates the exact fault currents for a three-phase fault at $t = 0.1$ s and trip of R7 at $t = 0.27$ s. The figure also demonstrates transient states and an increase in fault currents in the pertinent relays. Conventionally, by ignoring the FCR, the primary or secondary fault current is applied to set and to coordinate the relays. While both need to be considered. FCR must be considered during relays coordination; otherwise, there will be a big error in computing OTs.

2.4. Disregarding backup relays chain

Typically, for a particular fault location, only one OT is assigned to each relay. Besides, the coordination is only examined in the relay chain of the first-main relay as a constraint on optimization. When FCR is considered, it is necessary to correct the OT expression of the backup relays. The backup chains of R1 and R8 are illustrated in Fig. 5. R9 exists in both chains and has two OTs: first, as the third backup of R1, when R1 fails to operate and second as the backup of R8 when R8 fails to operate. Consequently, for a particular fault in the L1 line, R9 has different OTs depending on the backup chain. Therefore, when FCR is considered, the OT of the first-main relay and its backup relays are different from the OT of the second-main relay and its backup relays. Disregarding the backup relay chain of the two main relays is the other weakness of relays conventional coordination.

3. The proposed method

DOCR coordination in the interconnected networks is solved as an optimization problem [19]. Analysis of the network in the steady state with the constant fault currents leads to the problems presented in Section 2. To overcome these problems it is proposed:

1. to use instantaneous fault currents instead of steady-state fault

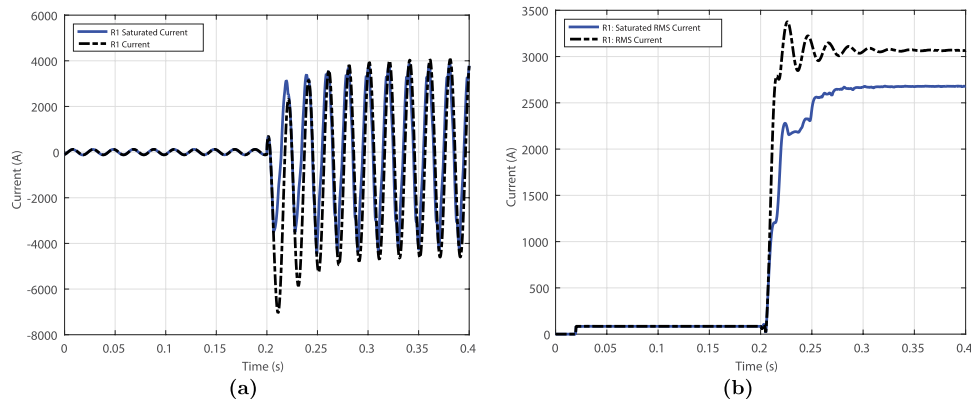


Fig. 3. CT saturation effect on instantaneous and effective currents of R1: (a) R1 instantaneous current; (b) R1 effective current.

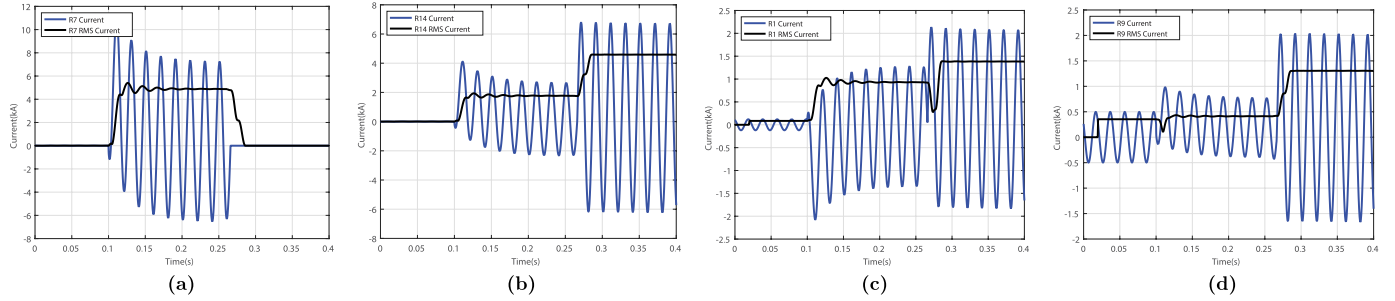


Fig. 4. Transient fault currents and their RMS for a fault near R7: (a) R7 currents as the first-main relay, (b) R14 currents as the second-main relay, (c) R1 currents as the backup of R14; (d) R9 currents as the backup of R14.

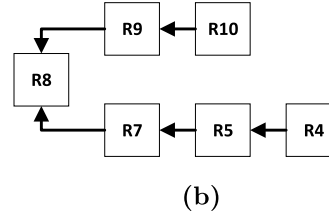
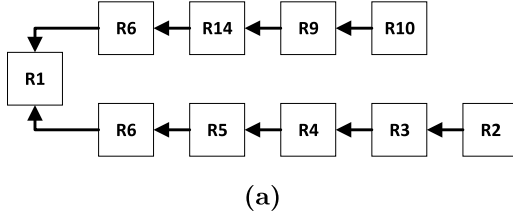


Fig. 5. Backup relays, (a) Backup chain of R1, (b) Backup chain of R8.

Table 1
Pseudo-code of the conventional methods.

Step	Code
1	Adjust the parameters of the optimization algorithm
2	Create the first population
3	while stopping criteria not met do
4	for each particle do
5	Generate a new position for the particle (relays new settings)
6	for each fault location calculate operating times based on steady-state fault currents
7	Calculate violations, then objective function
8	end do
9	Select the global best particle
10	end while

- currents by simulating the network in a time-domain software
2. to obtain actual OTs using instantaneous fault currents and simulate the relays precisely according to IEC standard
3. simulate the network three times in each iteration to let the fault currents redistribute.

To explain the proposed method and its difference with the conventional method, first, the pseudo-code of the common method and then the pseudo-code of the proposed method are presented.

The pseudo-code of the conventional method is presented in Table 1. In this method, initially fault currents are calculated only once in the steady state. Then, the OTs of relays are calculated using these currents in step 6 based on (1). Except for the key difference between the conventional and the proposed method in step 6, the other steps are similar.

The pseudo-code of step 6 in the proposed method is presented in Table 2. In step 6, the complete models available in Simulink are used

Table 2
Pseudo-code of step 6 in the proposed method.

Step	Code
6	for each fault location A. Simulate the network; disable the second-main relay and obtain the operating time of the first-main relay B. Simulate the network; disable the first-main relay and obtain the operating time of the backup relays related to the first-main relay C. Simulate the network; enable the first-main relay and obtain the operating time of the second-main relay and its backup relays

for circuit breakers, power transformers, synchronous generators, lines, and loads. CTs are modeled separately with the blocks available in Simulink [17]. Consequently, their effects on transient fault currents are exerted. DOCRs are simulated dynamically and exactly based on the IEC standard. Therefore, the effects of any transient signals on OTs are considered. Although these models are not exactly equivalent to the physical devices, their use is much better than simple conventional models.

To explain step 6 in more detail, let us consider Fig. 6. In this figure, the OTs of the DOCRs specified by arrow marks are obtained by simulation. As shown in Fig. 6a, the OT of Ra for the F1 fault is obtained by simulation in stage A of step 6 when Rb is inactive. Fig. 6b shows the stage B of step 6. The OT of Rc as the backup of the first-main relay (Ra) for F1 fault is obtained through the simulation when Ra is inactive and Rb is enabled to allow the occurrence of FCR. As shown in Fig. 6c, the OT of Ra as the second-main relay and its backup (Rc) for the F2 fault are obtained through the simulation in stage C of step 6. Here Rb is enabled letting the FCR happen. The OTs of relays in stages A and B are listed in the First-Main Relays Operating Times (FMROT) table and the OTs of relays in stage C are listed in Second-Main Relays Operating Times (SMROT) table.

4. Formulation of the coordination problem

The main purpose of the relay coordination is to determine the settings and to select the relay characteristic curve. TMS and PMS settings are the two main circumstances. The relay coordination is an optimization problem [19]:

4.1. Objective function

The settings of the main relays should be set so that the total OT of

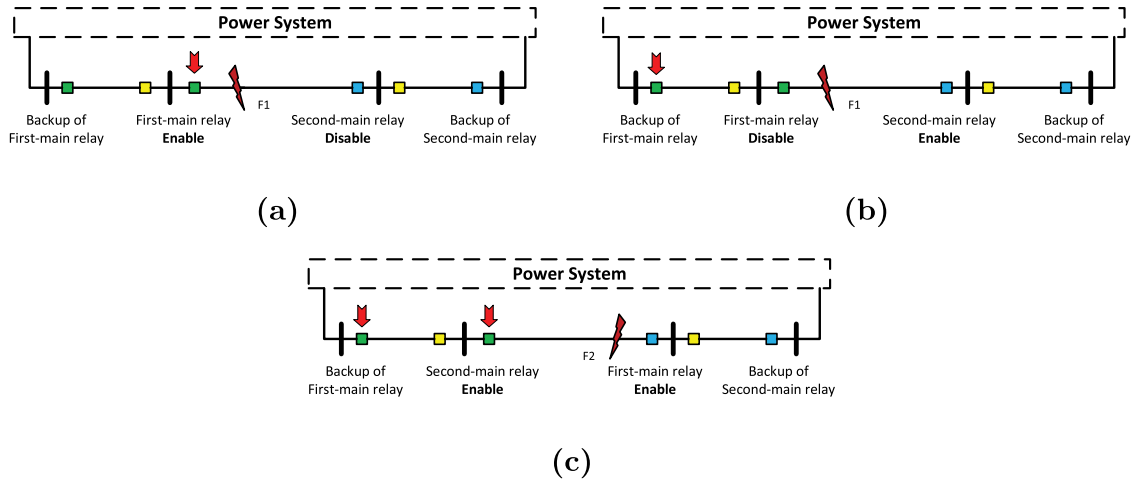


Fig. 6. Stage A, B and C of the step 6 in more detail: (a) Network to obtain the OT of first-main relay; (b) Network to obtain the backup OT of the first-main relay; (c) Network to obtain the OT of second-main relay and its backups.

Table 3
Constants of relay characteristic curve.

Curve type	α	k	c
Standard inverse (I)	0.02	0.14	0
Very inverse (V.I)	1	13.5	0
Extremely inverse (E.I)	2	80	0
Long inverse (L.I)	1	120	0

Table 4
Settings of relays in conventional method—case 1.

Relay	TMS	PMS	Curve type	Relay	TMS	PMS	Curve type
1	0.05	1.50	E.I	8	0.80	1.02	E.I
2	1.04	1.38	E.I	9	0.24	1.35	E.I
3	0.58	2.04	E.I	10	0.26	1.74	E.I
4	0.39	1.78	E.I	11	0.89	1.26	E.I
5	0.08	1.57	E.I	12	0.70	1.84	E.I
6	0.54	1.31	E.I	13	0.84	0.42	E.I
7	0.92	1.11	E.I	14	0.45	1.63	E.I

these relays is minimized for near and far faults. Accordingly, the objective function of the coordination optimization problem is as follows [5,6,20]:

Table 5
Operating time of relays in conventional method—case 1.

Near end relay #	Relay number													
	1	2	3	4	5	6	7	8	9	10	11	12	13	14
1	0.06					0.46		0.48						
2	0.72	0.32				5.35	0.72		0.52	2.19				
3		0.88	0.48				4.54			0.53				
4		2.37	1.32	0.35							1.57	2.63		
5				0.86	0.14							0.98	4.25	4.27
6				3.16	0.53	0.13		4.76					0.79	0.79
7				3.16	0.53		0.1	4.76					0.79	0.79
8	0.72					5.35	0.72	0.12	0.52	2.19				
9		0.89					4.54		0.13	0.53				
10		2.37	1.32						0.22	1.57	2.63			
11				0.86						0.58	0.98	4.25	4.27	
12				3.15	0.53			4.76			0.39	0.79	0.79	
13						0.46		0.48				0.08		
14	0.72					5.35	0.72		0.52	2.19				0.1

$$\text{Objective Function} = \sum_{i=1}^{N_{\text{near}}} t_{\text{nearmain}}^i + \sum_{j=1}^{N_{\text{far}}} t_{\text{farmain}}^j \quad (4)$$

where, t_{nearmain} and t_{farmain} are the OT of the main relay, respectively. Also, N_{near} and N_{far} are the numbers of relays for near and far end three-phase faults, respectively.

4.2. Coordination criteria

The coordination limit between the OT of relay pairs is as follows:

$$t_b - t_m \geq \text{CTI} \quad (5)$$

where, t_b and t_m are the OT of backup and main relays respectively. This limitation aims at creating a Coordination Time Interval (CTI) between relay pairs.

4.3. Relay characteristics

In this paper, the Inverse Definite Minimum Time (IDMT) relay is simulated in Simulink based on (2) and (3) with no filter in the inputs [21]. The line three-phase voltage signal is used to simulate the directional function of the relay and its polarization. The constant values in (2) depend on the characteristic curve according to Table 3.

Table 6
Actual operating time of first-main relays and their backups in conventional method (FMROT)—case 1.

Near end relay #	Relay number													
	1	2	3	4	5	6	7	8	9	10	11	12	13	14
1	0.07					0.52								1.31
2	0.67	0.34				3.22	0.68							
3		0.77	0.5			5.89	1.42							
4		1.97	1.45	0.37			4.24							
5		2.41	1.8	0.97	0.16		4.9							
6				2.83	0.91	0.14								0.88
7				2.2	0.82		0.11	1.78					0.86	
8							0.8	0.13	0.87	2.06				
9									0.15	0.61	1.94	2.59		5.2
10										0.23	1.53	1.96	5.36	4.12
11											0.6	0.99	2.27	1.53
12								2.89				0.41	0.71	0.71
13							1.25	0.52		5.57			0.1	
14	0.75					1.83			0.79	1.61				0.12

Table 7
Actual operating time of second-main relays and their backups in conventional method (SMROT)—case 1.

Near end relay #	Relay number													
	1	2	3	4	5	6	7	8	9	10	11	12	13	14
1							0.86	0.26		4.61				
2								0.41	0.65	2.04	2.91	7.64	6.8	
3									0.55	1.59	2.23		5.18	
4								6.91		0.76	1.06		1.62	
5								4.72			0.61	1.58	1.06	
6							1.31	0.45				0.23		
7	0.5					2.35				2.58			0.25	
8	0.21					0.46							1.41	
9		0.55				5.78	1.14							
10		0.87	0.59			8.36	1.66							
11		2.09	1.47	0.75			5.05							
12		2.53	1.7	0.87	0.48		5.97							
13				4.64		0.27							0.91	
14				3.25			0.25	2.16				0.56		

Table 8
Operating times of relays for a fault near R8—case 1.

Fault location	Method	Relay number			
		7	8	9	10
8	Conventional (row 8)	0.72	0.12	0.52	2.19
8	FMROT (row 8)	0.80	0.13	0.87	2.06
1	Conventional (row 1)		0.48		
1	SMROT (row 1)	0.86	0.26		4.61

4.4. Constraints on PMS and TMS

Relay settings have some limitations; the time and current settings cannot be set to any desired value. The lower limit of *PMS* is 1.5 times the maximum load current and upper limit is 2/3 times the minimum

Table 10
Miscoordinations in conventional method—case 1.

FMROT			SMROT					
Near end relay	Main	Backup	Near end relay	Main	Backup	Near end relay	Main	Backup
2	2	1	8	1	6	1	9	10
2	2	7	9	1	6	2	9	10
3	3	2	10	3	2	4	11	12
3	1	6	10	1	6	4	13	8
11	11	12	12	5	4	6	13	8
12	12	14	13	5	4	7	14	1
12	12	13	14	7	13	7	9	10
13	9	10	14	5	4			

Table 9
Percent error in the first and second main relays OTs in conventional method—case 1.

Relay number	1	2	3	4	5	6	7	8	9	10	11	12	13	14
First-main relay	−15	−3	−3	−4	−7	−7	−13	−9	−10	−5	−2	−3	−11	−11
Second-main relay	128	−6	−10	109	106	191	221	181	117	139	13	−13	102	192

Table 11
Expectable second-main relays without backup.

Near end relay number	Main relay	Backup relay
9	2	1
13	6	5
14	7	5
1	8	9
7	14	9

Table 12
Settings of relays in proposed method—case 1.

Relay	TMS	PMS	Curve type	Relay	TMS	PMS	Curve type
1	0.39	1.09	E.I	8	0.9	2.66	E.I
2	0.65	2.8	E.I	9	0.08	3.21	E.I
3	0.39	3.18	E.I	10	0.58	2.37	E.I
4	0.83	2.09	E.I	11	0.64	1.7	E.I
5	1.09	0.88	E.I	12	0.6	2.57	E.I
6	0.23	2.85	E.I	13	0.27	2.12	E.I
7	0.54	2.71	E.I	14	0.23	3.54	E.I

three-phase fault current [1,22]. These limitations appear in the problem of optimization as follows:

$$TMS_i^{\min} \leq TMS_i \leq TMS_i^{\max} \quad (6)$$

$$PMS_i^{\min} \leq PMS_i \leq PMS_i^{\max} \quad (7)$$

where, $TMS_i^{\min}$ and $TMS_i^{\max}$ indicate the lower and upper limits of the TMS , respectively, and $PMS_i^{\min}$ and $PMS_i^{\max}$ represent the lower and upper limits of the PMS for i th relay, respectively.

4.5. Parameters value

In this paper, the following values are selected for the optimization problem [23,24]:

$$TMS \in [0.05 - 1.1], \quad \text{step} = 0.01 \quad (8)$$

$$CTI = 0.4 \quad (9)$$

5. Simulations and results

The proposed method is evaluated in three case studies. In the first case, the IEEE 8-bus network is coordinated with the common method and the miscoordinations are depicted. The network is then coordinated

with the proposed method and it is shown that all the miscoordinations are removed. In the second case study, the settings of the state-of-the-art paper are evaluated by the proposed method. In the third case, the same steps of the first case are repeated in the most commonly used network; i.e., the IEEE 30-bus network.

5.1. Case study 1: IEEE 8-bus network

For the first case study, the IEEE 8-bus network (Fig. 1) is examined. Several researchers have used this network to evaluate their methods. In the following, this case study is explained in full details.

5.1.1. Coordination based on conventional method

The network was coordinated based on the conventional method. The objective function was 14.4 s, and the settings of relays were obtained according to Table 4. With these settings, the OT of the relays conforms to Table 5. In this table, the rows and columns depict the near end relay number and the relay OT. The null space indicates that the relay has no operation up to 9 s after the fault. The main relay is marked with a solid border cell and its backup relays are depicted with a dashed line border cell in each row. To explain the table, the fault near R1 in the L1 line is checked out. The OT of R1 is 0.06 s for a near fault. The OT of its backup relay (R6) and the second-main relay (R8) are 0.46 s and 0.48 s, respectively.

To investigate the quality of the relays settings obtained by the common method, the network was simulated meticulously in the simulation software. As a result, all factors included in Section 2 were considered. Afterward, the relays OTs were obtained from the model for all faults. Tables 6 and 7 show the FMROT and SMROT tables, respectively. The OT of R9 is examined for a fault near R8. According to Table 5, the OT of R9 is 0.52 s. However, it is 0.87 s (Table 6, row 8) as the backup of R8 and has no operation as the backup relay in the R1 chain (Table 7, row 1). Comparing the OTs in Tables 5–7, two problems are identified:

1. If the network is simulated, the relays will have different OTs.
2. The conventional method does not detect 23 miscoordinations.

Table 8 shows the OTs of R8 and its backups as the first and second main relays. The OT of R8 as the first and second main relays were corrected from 0.12 s to 0.13 s and from 0.48 s to 0.26 s, respectively. The OTs of the backup relays were corrected, as well. For instance, the OT of R7 was corrected from 0.72 s to 0.8 s. Similar changes occur in the OT of other relays, as well.

Table 13
Operating time of first-main relays and their backups in proposed method (FMROT)—case 1.

Near end relay #	Relay number													
	1	2	3	4	5	6	7	8	9	10	11	12	13	14
1	0.24					1.09								5.81
2	2.55	0.86				8.73	2.41							
3		2.4	0.82				4.85							
4		3.89	2.28	1.08										
5		5.65	3.07	2.13	0.61									
6				9.85	2.7	0.28								2.2
7				6.37	2.21		0.36						3.54	
8							2.79	0.93	1.94	8.38				
9									0.27	2.32	3.67	5.24		
10										0.93	2.19	3.34		
11											0.79	1.75	4.39	
12												0.68	5.52	2.19
13								3.78	2.05				0.74	
14	2.88					5.1			1.93	6.49				0.27

Table 14
Operating time of second-main relays and their backups in proposed method (SMROT)—case 1.

Near end relay #	Relay number													
	1	2	3	4	5	6	7	8	9	10	11	12	13	14
1							3.09	1.55						
2									0.99	2.13	3.16	5.45		
3										1.43	2.23	3.82		
4											1.41	2.11		4.33
5												1.19		2.85
6							4.77	2.39					0.7	
7						6.01								0.62
8							1.61							4.51
9								3.64						
10									5.76					
11														
12														
13														
14														

Table 15
Miscoordinations in Sharma and Panigrahi [7]—case 2.

FMROT						SMROT					
Near end relay #	Main	Backup	Near end relay #	Main	Backup	Near end relay #	Main	Backup	Near end relay #	Main	Backup
1	5	4	6	5	4	9	2	7	2	11	12
2	2	1	7	5	4	9	1	6	3	11	12
2	2	7	8	8	7	10	1	6	4	11	12
2	5	4	9	11	12	11	4	3	4	13	8
3	2	7	10	10	11	11	2	7	6	8	7
3	1	6	10	11	12	11	1	6	7	9	10
4	4	3	11	11	12	12	5	4			
4	1	6	12	12	14	12	4	3			
5	5	4	12	12	13	12	2	7			
5	4	3	12	9	10	12	2	1			
5	2	7	13	9	10	13	5	4			
5	2	1	14	14	9	14	5	4			
6	6	14				1	9	10			

Table 9 presents the percent error in the first and second main relay OTs. The maximum first-main relay OT error is -15% (R1). This error is the result of transient state ignorance. The maximum second-main relay OT error is 221% , which belongs to R7. This error is mainly

induced by ignoring the FCR.

Based on the R1 backup chain, R10 is the backup of R9. According to Table 7, the OT of R10 is 4.61 s, where R9 has no operation, and this is a miscoordination. Table 10 shows all miscoordinations generated in the conventional method.

According to Table 7, 5 second-main relays remained without a backup. Table 11 lists these relays. Because, when the first-main relay breaks the network, the direction of the power flow varies at the backup relays due to the FCR, and as a result, the relays are not polarized.

5.1.2. Coordination based on the proposed method

To examine the proposed method, the same network with the same constraints was coordinated under the pseudo-code of the proposed method. In other words, by considering the factors in Section 2, the complete network model and the DOCRs were applied to obtain the transient fault currents and the relays' OTs. The objective function was 24.65 s. Table 12 shows the settings of the relays based on the proposed method. Tables 13 and 14 present FMROT and SMROT tables using these settings, respectively. Evaluation of the OTs indicates no miscoordination. Since the current detected by the relays changes over time, the Time Current Curve (TCC) to present the coordination between relay pairs is not appropriate.

As expected, like the conventional method, some second-main relays have been left without a backup due to the changing of the power flow direction. These DOCRs are listed in Table 11.

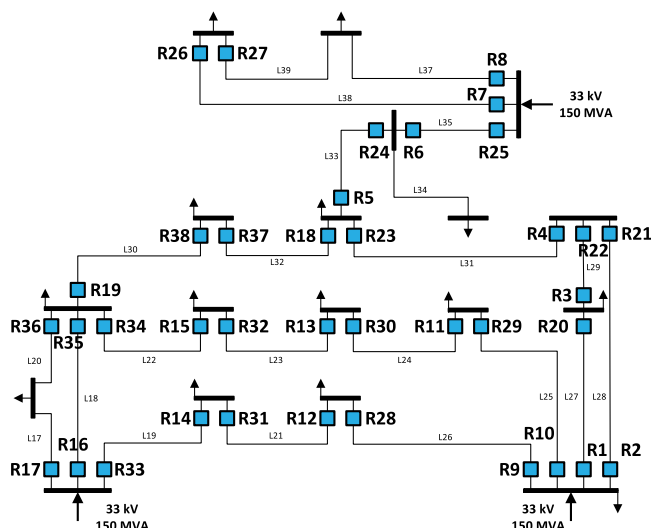


Fig. 7. Distribution part of IEEE 30-bus network.

Table 16
Settings of relays for IEEE 30-bus network—case 3.

Relay	Conventional		Proposed		Relay	Conventional		Proposed	
	TMS	PMS	TMS	PMS		TMS	PMS	TMS	PMS
1	0.19	2.288	0.47	1.998	20	0.08	1.001	0.06	2.03
2	0.06	2.442	0.56	0.636	21	0.13	0.505	0.12	1.261
3	0.11	2.699	0.66	0.669	22	0.17	1.101	0.2	2.013
4	0.19	1.912	0.17	1.648	23	0.11	1.491	0.07	3.047
5	0.06	2.981	0.1	2.523	24	0.23	0.896	0.08	2.245
6	0.11	1.378	0.14	1.529	25	0.13	2.225	0.1	2.501
7	0.1	2.249	0.33	0.697	26	0.66	0.617	0.1	0.642
8	0.11	1.591	0.08	1.107	27	0.53	0.968	0.3	0.301
9	0.2	2.301	0.38	2.738	28	0.23	0.616	0.23	1.111
10	0.1	3.658	0.11	4.375	29	0.08	0.849	0.08	1.125
11	0.3	0.832	0.26	1.69	30	0.17	0.766	0.17	1.13
12	0.2	1.235	0.18	2.531	31	0.12	1.62	0.27	1.383
13	0.09	2.072	0.2	1.489	32	0.25	0.8	0.2	1.406
14	0.14	1.136	0.17	2.127	33	0.22	1.363	0.26	1.908
15	0.15	0.907	0.14	1.414	34	0.36	0.743	0.27	1.518
16	0.14	2.441	0.14	2.967	35	0.16	1.007	0.16	1.546
17	0.24	0.609	0.07	1.864	36	0.11	0.33	0.26	0.483
18	0.29	1.054	0.12	2.125	37	0.25	0.468	0.18	1.208
19	0.38	0.489	0.22	1.529	38	0.41	0.323	0.1	1.643

5.2. Case study 2: evaluation of the state-of-the-art paper

For the second case, the results in Sharma and Panigrahi [7] were evaluated as a state-of-the-art paper by the proposed method. According to the proposed method, the FMROT and SMROT tables for the result of this paper were obtained by simulation. Examination of the OTs based on these tables shows 44 miscoordinations. Table 15 presents all these miscoordinations.

5.3. Case study 3: IEEE 30-bus network

The third test system is the distribution part of the IEEE 30-bus

network (Fig. 7) to evaluate the performance of the proposed method in large cases [25]. For simplicity, it is assumed that all DOCRs have the standard inverse characteristic curve. Table 16 presents the optimal settings of relays in the case the coordination is done using conventional and the proposed method. A total of 94 miscoordinations are not detectable by the conventional method. The miscoordinations related to main relays and their backups are listed in Table 17. For the sake of brevity, FMROT and SMROT tables, which have 38 rows and columns, were omitted from the paper. Table 18 shows results similar to Table 9 for this network. As expected, all miscoordinations were eliminated in the proposed method. Furthermore, the obtained results emphasize that the proposed method is an efficient algorithm for coordination and

Table 17
Important miscoordinations in conventional method—case 3.

FMROT						SMROT					
Near end relay #	Main	Backup	Near end relay #	Main	Backup	Near end relay #	Main	Backup	Near end relay #	Main	Backup
3	3	1	28	28	31	1	20	22	20	1	21
4	4	3	31	31	33	9	28	31	21	2	20
9	9	29	33	33	35	10	29	30	22	3	1
9	9	21	34	34	16	11	30	32	24	5	37
11	11	10	34	34	38	12	31	33	32	13	11
18	18	4	38	38	18	13	32	34	35	16	36

Table 18
Percent error in the first and second main relays OTs in conventional method—case 3.

Relay #	FMROT	SMROT	Relay #	FMROT	SMROT	Relay #	FMROT	SMROT
1	−1.2	40.3	14	−1.5	6.4	27	−2.7	31.3
2	−3.9	149.9	15	−6.6	4.7	28	0	1.3
3	−3.1	0	16	−4	22.8	29	−6.4	19.5
4	0	5.7	17	−1.8	62.8	30	−5.2	3.2
5	−4.2	−0.3	18	−5	0.5	31	−1.3	5
6	−2.8	−2.2	19	−2.1	7.4	32	−4.1	5.7
7	−3.7	NO ^a	20	−2.9	29.9	33	−3.8	13.5
8	−4.6	NO	21	−6.1	139.6	34	−1.9	10.6
9	−3.2	9.5	22	−4.6	32.9	35	−3.2	27
10	−1.6	58.4	23	−1.6	11.3	36	−4.3	128.9
11	−5.6	0	24	−5.7	0.1	37	−4.1	8.9
12	−2.6	22.9	25	−5.1	0	38	−3.5	0
13	−4	6.3	26	−0.9	9.1			

^a No operation.

considering the transient state of the network in the large case power systems.

6. Conclusion

In conventional methods, the relays are coordinated in a steady state. The common formula for calculating the operating time of relays is based on stable fault currents. Due to the common use of the simplified model of network elements, the operating time (OT) of the relays is not accurately calculated. The error in the calculation of the operating time of the relays results in miscoordinations in practice.

To deal with this issue, it is proposed using instantaneous fault currents instead of steady-state fault currents by simulating the network and relays in a time-domain software during coordination. Therefore, it was possible to use a complete model of network elements.

The results of this study showed that the OT of backup relays has not been properly stated in previous papers. A new technique was proposed to show the backup relays OTs while the redistribution of fault currents is considered. Network simulation was performed three times in each iteration to consider the redistribution of fault currents. Accordingly, FMROT and SMROT tables were introduced. Using the proposed method, 23 and 94 miscoordinations as the result of the conventional method of coordination in the IEEE 8-bus and IEEE 30-bus networks were eliminated.

During the optimal coordination, the network simulation in the time domain prevents miscoordinations. The simulation also provides relays exact OTs. The proposed method employs relays the exact formula and considers the following factors to prevent miscoordination: transient state of the network through the simulation and using the complete model of the elements, topology transient change (fault current redistribution), and backup relays chain. Overall, error-free OTs are obtained by considering transient states and consequently miscoordinations are prevented.

Declaration of Competing Interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Afshin Mahboubkhah: Conceptualization, Methodology, Software, Formal analysis, Writing - original draft, Writing - review & editing, Visualization. **Vahid Talavat:** Investigation, Methodology, Validation, Supervision, Writing - review & editing. **Mojtaba Beiraghi:** Writing - original draft, Supervision, Writing - review & editing.

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